Datasheet

NVIDIA OVX

Purpose-built for powering the creation and operation of Omniverse applications at data center-scale.

Virtual worlds and digital twins—physically accurate, AI-enabled simulations perfectly synchronized with the real world—will enable the next era of AI and industry, accelerating innovation by allowing enterprises to design, simulate, and optimize products, equipment, and processes in real-time—before ever going into production.

Large-scale 3D virtual worlds and digital twin simulations must operate with full-design fidelity in real time and be perfectly synchronized. Building and operating these complex and immersive environments requires a new computing system—NVIDIA OVX™.

Powerful Performance for Omniverse In The Data Center

NVIDIA OVX is a computing system reference architecture purpose-built for powering the creation and operation of NVIDIA Omniverse™ applications at data center-scale. NVIDIA OVX servers, based on this reference architecture, deliver a powerful combination of RTX-accelerated graphics and AI capabilities and high-performance networking to deliver breakthrough performance for the most demanding Omniverse workloads like true-to-reality simulations, immersive 3D virtual worlds, and digital twins..

Validated Architecture for Omniverse at Scale

Each component within the OVX architecture has been selected, tested, and validated to provide the computing performance required for building and operating physically accurate, AI-enabled virtual simulations perfectly synchronized with the real world.

Each NVIDIA OVX server features 4 NVIDIA L40 GPUs, based on the Ada Lovelace architecture, that deliver next-generation graphics and AI capabilities to accelerate Omniverse workflows, including ray-traced and path-traced rendering of materials, physics simulation, and the generation of physically accurate 3D data for perception network training. The L40 GPUs are combined with NVIDIA ConnectX®-7 network adapters, a BlueField-3 DPU, top-of-line CPUs, and super-fast NVMe storage in leading data center platforms available from NVIDIA-Certified partners.

NVIDIA OVX Server

Hardware Specifications	
GPU	4x NVIDIA L40 GPUs
GPU Memory	192GB total
CPU	2x 32 Cores
System Memory	1TB
Storage	8TB NVMe
Networking	NVIDIA ConnectX-7 NVIDIA BlueField-3

Germany's national infrastructure manager is designing an AI-enabled digital twin, powered by Omniverse and OVX, to develop the future railway system.

Unparalleled Performance and Scalability

The NVIDIA OVX hardware architecture is designed to expand to SuperPOD-scale and delivers a fully-integrated platform for the most complex Omniverse Enterprise workloads. The NVIDIA OVX SuperPOD architecture offers scalable performance for operating real-time simulations and AI-enabled digital twins at factory, city, or planetary scale. The building block of the OVX SuperPOD is a scalable unit that includes 64 OVX servers, high-performance compute and storage platforms, and networking infrastructure based on the NVIDIA Spectrum-3 ethernet switch system.

Build and Operate 3D Virtual Worlds with Omniverse

NVIDIA Omniverse Enterprise is a scalable, end-to-end platform enabling enterprises to build and operate metaverse applications. Based on Universal Scene Description (USD), Omniverse Enterprise allows teams to connect and customize complex 3D pipelines and operate large-scale, physically accurate virtual worlds.

NVIDIA Omniverse Enterprise deployed on NVIDIA® OVX™ delivers a scalable, integrated hardware and software platform to build and operate virtual worlds and accelerate workflows at any scale. OVX provides groundbreaking RTX-accelerated visualization, compute, and state-of-the-art networking that allows workloads to scale across 100s of nodes. From building custom 3D pipelines to operating large-scale, physically accurate, AI-enabled virtual simulations, NVIDIA Omniverse Enterprise and OVX can accelerate the most complex industrial, scientific, and AI workflows associated with large-scale modeling and simulation.

NVIDIA cuOpt and Isaac Sim, powered by Omniverse and OVX, accelerate logistics workflows like factory layout optimization and AMR fleet training.

NVIDIA OVX Availability

NVIDIA OVX systems are available globally through the following partners:

DELL Technologies · GIGABYTE™ · Hewlett Packard Enterprise · Lenovo · QCT™ · SUPERMICRO®

Ready to Get Started?

To learn more about the NVIDIA OVX, visit:
www.nvidia.com/en-us/omniverse/platform/ovx

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